

Lecture 18

Tuesday April 14th

Quiz on the 20th

Deadlocks Recap

- mutual exclusion
- hold and wait
- circular wait
- Resource allocation graph
- cycles in the graph lead to deadlock

Deadlock prediction and Prevention

- preemption
- circular wait
- single resource
 - detection relies on creating the graph and detect a cycle
 - use an algorithm for cycle detection
- Multiple resource
 - manual is too much work so use algorithm detection
 - cycle detection will not work
 - Allocation matrix
 - rows are processes (allocation, Request, Available, Work) - work is a copy of available
 - columns are resource types
 - if a single sequence exists that can execute you will not have a deadlock

Deadlock Recovery - Process Termination

- Abort all deadlocked processes
- abort one process at a time
- How to choose which process gets terminated
 - Priority of process
 - How long has process has left to run
 - resources the process has used
 - resources needed to complete

- How many processes will be terminated

Deadlock Recovery - Resources Preemption

- selecting a victim - minimize cost
- Rollback - return to a safe point
- Starvation

Deadlock Avoidance

- Requires that the system has some additional a priori information available
 - request disk then printer
 - or request at most n resources
 - allows for more concurrency

Safe state

- system is in a safe state if there are no deadlocks and all process can run
 - NO DEADLOCKS
- Avoidance ensures that we will not enter a deadlocked state
- Deadlocks only exist in the unsafe state
- Example
 - 12 disks with 3 processes
 - 9 are currently running so 3 available processes
 - p2 needs 4 and already has two so only needs 2 more will release when done so available will go up to 5
 - p1 needs 10 and has 5 with 5 being available so it can run then resource
 - p3 has 9 and needs 2 more so it can run then release for 12 available at the end
 - sequence is [p2, p1, p3]
 - no deadlock

Banker's Algorithm

- Works for multiple resource instances
- Each process declares max # of resources it may need
- When a process requests a resource, it may have to wait if this leads to an unsafe state
- When a process gets all its resources it must return them in a finite amount of time
- Resource-Request algorithm